

CHALLENGE TITLE

Down to the Meter

How might we find and build what only becomes possible when positioning comes down to one meter?

PARTNER

EWIA.tech, an ESA BIC Hessen incubatee, is bringing high-precision satellite-based localization to the mass market of low-cost receivers, including those used in mobile phones.

THE PROBLEM

For decades, satellite positioning meant “a few meters off, more between buildings.” That era is ending. Dual-frequency receivers and free correction services such as Galileo’s High Accuracy Service are bringing positioning down to around one meter on ordinary hardware, with no base stations and no extra infrastructure.

Here is the strange part: most of the digital world will never notice. Maps, food delivery and ride hailing are perfectly happy at five meters. But scattered across industries are people for whom exactly those few meters of error quietly break things today: a service that cannot get approved, a fine that cannot be contested, a safety call that cannot be trusted, a product that ships with a disclaimer instead of a promise.

The trap is thinking from the technology forward: “one meter exists, what app could we build?” That road produces demos nobody needs. The discipline is thinking from the user backward: find someone whose documented pain exists because positioning is too coarse, and prove that one meter flips their situation. Companies building precise positioning, EWIA among them, face this exact question every week. This weekend, you do the finding and the proving.

THE OPPORTUNITY

Positioning precision is becoming a commodity input, like cloud compute a decade ago. When that happens, the value moves from the capability to the applications, and the teams who find an underserved vertical early get to own it.

A few starting points that pass our filter, any one a fair place to begin:

- **Drones in tight airspace.** Approval buffers scale with position uncertainty; at one meter, corridors and missions become approvable that are impossible today.
- **Rail work safety.** Parallel tracks lie four meters apart. At five meters GPS cannot tell which track you are standing on; at one meter it can.
- **Navigation for blind and low-vision people.** At five meters you are “somewhere on the street,” at one meter you are at the right crossing, the right bus door. Microsoft built this (Soundscape), shut it down in 2023 and open-sourced it.
- **Micromobility compliance.** Sidewalk, bike lane and street lie two to four meters apart. Operators cannot prove where a scooter was parked, and cities fine them for it.
- **Sailing race starts.** The start line is invisible on water and over-early calls are made by eye. One meter calls it.
- **Or your own.** The list is a runway, not a fence. The best outcome of this challenge is a use case nobody in the room has thought of.

One thing keeps this challenge honest: accuracy is the variable, not the assumption. The convincing demos show concretely what flips between today's roughly five meters and tomorrow's one. Build that flip and show it.

CONSTRAINTS

- The necessity test is a hard boundary: your use case must demonstrably fail at today's accuracy and work at about one meter. If five meters is good enough, it is the wrong use case for this challenge.
- Outdoor scenarios only. Satellite navigation does not work indoors.
- Safety- or compliance-critical claims must trace back to data and stated assumptions. No black-box trust.
- Use real geodata for your chosen domain: real maps, real zones, real track layouts. Positions may be simulated; the world may not.

DELIVERABLES

Standard LAUNCH Build Days deliverables:

1. A sharp customer and problem, the heart of this challenge: who exactly is blocked by coarse positioning today, and what public evidence (incident reports, fines, regulations, market data) shows the pain is real. Desk research is enough, no interviews expected over a weekend.
2. A solution concept with a clear value proposition: what you are building, for whom, and why it beats how this is done today.
3. A working tool on real geodata that takes a position track, treats each fix as a point plus an error radius (about ten meters for a standard receiver, about two for a corrected one), and computes the operator's decision rather than just drawing it: inside a no-fly zone, which of two tracks, did the boat cross the line, the bike lane or the street. Then it shows how that decision flips between five meters and one meter, a toggle is the clearest way. Simulated or replayed positions are expected; the world must be real. A narrow, deep slice beats a broad, shallow demo.
4. An Expo pitch: five minutes to present and demo, then four minutes of jury Q&A. Show the working thing, not slides.

EVALUATION CRITERIA

Criterion	Description
Problem Validation	A concrete user and a real pain, grounded in public evidence. Desk research is enough; no interviews expected. "Drones are the future" is not a problem.
Solution Quality	The use case is well justified against the necessity test, and the tool genuinely uses the accuracy: the decision depends on the meter, not on a perfect-position assumption.
Prototype Tangibility	A working tool that computes a real decision from position data on real geodata, not slides and not just a map. A narrow, deep slice done well beats a broad, shallow demo.
Business Potential	Could this become a real venture? Is there a market and a model beyond a single customer?

Criterion	Description
Pitch and Team	Clear and confident. The team understands both the user's world and what positioning can and cannot do.

Partner bonus (EWIA): the best treatment of position uncertainty as a variable, showing concretely what changes between five meters and one meter. Graceful degradation counts just as much, what the tool does at two meters or when the fix drops, instead of assuming a perfect position.

RESOURCES

Start here - The starter floor, in the kit: `geo_core.py` for the geometry (point in polygon, distance to an edge or a line, nearest of several lines, a buffer test), `viewer.html` for a map with a five-meter / one-meter toggle, and `positioning_primer.md` on what the meter actually buys you. The plumbing is done so you build the decision logic. - Real GNSS error, in the kit: `tracks/real/zurich_agz_groundtruth.csv` with `real_data.py`, a real drone's on-board GPS against ground truth (RMS about four meters). Replay real coarse error onto your own route. - `generate_tracks.py`, in the kit: your waypoints in, a truth track plus a five-meter and a one-meter track out, for whatever route you pick.

Geodata for your domain (you source it; the world must be real) - OpenStreetMap via Overpass, overpass-turbo.eu: rail, sidewalks and bike lanes, harbors, crossings, airfield layouts. The most useful source for almost any vertical. For airspace: dipul and OpenAIP. - Two traps that quietly break the flip: parallel rail tracks are often one way tagged `tracks=2`, not separate geometries; bike lanes are often a `cycleway` tag on the road, not offset geometry. Pick a place mapped as separate ways, or draw the lines yourself.

More position data - EWIA will generate five-meter and one-meter tracks fitted to your scenario on request; bring your route to the Monday-evening Q&A. - Google Smartphone Decimeter Challenge on Kaggle: real raw smartphone GNSS with precise ground truth, if you want a second real-error source. Needs a free Kaggle account.

Inspiration (reference, not to copy) - Droniq (drones), Vakaros (sailing lines), ForeFlight (pilots): each is a vertical that already monetized position.